11-04 第1篇

阅读理解（说明文） Why Do Onions Make Us Cry?

① Any student who has ever helped Mum slice an onion on a Saturday morning knows the scene: the knife comes down, the onion opens, and **half a minute later** the kitchen is full of quiet tears. The tears are real. The reason, however, is not sadness. Onions do not make us feel anything emotional. They simply **win a small war** against our eyes every time we cut them. To understand why, we have to meet one of the cleverest defence chemicals in any kitchen plant.

② Onions have lived in the wild for a long time, and in the wild they have one big problem: **small animals like to eat them**. A rabbit, a mouse, a hungry insect — the onion cannot run away, and its soft, wet leaves look like an easy meal. Over millions of years, onions have developed a clever chemical **self-defence** system. The trick is this: inside the onion's cells, two **special chemicals** are kept carefully **apart** from each other. While the onion is whole, the two never meet, and the onion tastes and smells mild.

③ But the moment the cells are **broken** — by a mouse's tooth, or your kitchen knife — the two chemicals suddenly mix. The first is a sulphur compound called **amino-acid sulphoxide**. The second is an **enzyme** that sits nearby, ready to react. In just **a few seconds**, these two mix to form a tiny gas called **propanethial-S-oxide**. This gas rises into the air in front of the chopping board.

④ Gases, of course, are invisible. But this one is very **attracted** to a certain place: the thin, wet film that covers your eyes, called the **tear film**. When the gas touches this wet film, it reacts with water to form a **mild acid**. The acid is far too weak to hurt the eye, but your eye has a very sensitive alarm system. It immediately sends a signal to the brain: "**Foreign thing on cornea! Rinse immediately!**"

⑤ The body's answer is already prepared: the tear glands above your eyes begin to **pump extra water** to wash the acid away. You blink. You rub your eye. You cry. Notice that the "crying" here is not emotional, and you are not sad at all. It is simply the body's quiet way of saying, "**Something slightly chemical is in the air; let us flush it out.**" In about one minute, if the onion is taken away, the tears usually stop.

⑥ Chefs and grandmothers, who cut onions every day, have come up with several small tricks to **slow this war**. The simplest is to use a **very sharp knife**. A sharp knife cuts fewer cells on each stroke, and fewer broken cells mean less gas. A dull knife, by contrast, **crushes** many more cells with each cut, and crushing sets off far more of the

chemical reaction. So a clean, well-sharpened knife is a gentler guest at the chopping board than a blunt one.

⑦ A second trick is **cold**. If you put the onion in the fridge for **15 to 30 minutes** before cutting, the enzyme works more slowly in the cold. Less gas rises; fewer tears fall. A third trick is to cut under **a little running water**, or near an open window where moving air carries the gas away from your face. A fourth, used in restaurants, is to wear a pair of **swimming goggles** — the gas simply cannot reach the tear film through the waterproof lens.

⑧ A final small question: why do some onions sting more than others? The answer is **the soil**. Onions grown in soil rich in sulphur — for example, some yellow onions from northern provinces — produce much more of the defence chemical. Their taste is stronger, and their tears are bigger. Sweet red onions or spring onions, grown in milder soil, cry much less. So the next time your eyes are running in a warm kitchen, remember: **you are not sad; you are, in a small way, being politely asked by a vegetable to leave it alone.** Smile, blink, and carry on — supper is almost ready.

注释： enzyme 酶；sulphur 硫；cornea 角膜；tear gland 泪腺；goggles 护目镜；crush 碾碎

根据短文内容，选择最佳答案。

1. Why have onions developed a chemical "self-defence" system?

A. Because they want humans to cry in the kitchen.
B. Because in the wild, small animals like rabbits, mice and insects like to eat them; the onion cannot run away.
C. Because they want to grow bigger.
D. Because they do not like cold weather.

2. How does cutting the onion trigger the formation of the stinging gas?

A. The knife adds a new chemical to the onion.
B. The broken cells allow two chemicals kept apart — a sulphur compound and a nearby enzyme — to mix and form propanethial-S-oxide.
C. The onion feels pain and releases heat.
D. Water from the onion makes a new substance by itself.

3. Why do our eyes "cry" when we cut an onion?

A. Because the gas makes us feel sad.
B. Because the gas reacts with water on the tear film to form a mild acid; the eye's alarm system asks the tear glands to flush it away.

C. Because onions contain salt.
D. Because the kitchen is too hot.

4. According to Paragraphs ⑥—⑦, which of the following is NOT a way to reduce onion-tears?

A. Using a very sharp knife so fewer cells are broken.
B. Chilling the onion in the fridge for 15—30 minutes before cutting.
C. Cutting under running water or near an open window.
D. Pressing hard on the onion with a dull knife to crush it quickly.

5. Why do some onions sting more than others?

A. Because of the cook's experience.
B. Because onions grown in soil rich in sulphur (such as some yellow onions from northern provinces) produce much more of the defence chemical, while sweet red onions and spring onions, grown in milder soil, sting less.
C. Because older onions always sting more.
D. Because of the kitchen temperature.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "in the wild ... small animals like to eat them ... cannot run away"
【解析】进化动机直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "two chemicals suddenly mix ... propanethial-S-oxide"
【解析】切割后酶与硫化合物反应生成刺激气体。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④⑤段 "reacts with water ... mild acid ... tear glands ... pump extra water"
【解析】物理化学反射流泪。A、C、D 均错。故选 B。

4.
【答案】D
【题型】细节理解题（否定）
【定位】第⑥⑦段"sharp knife / fridge / running water / goggles"
【解析】A、B、C 均在文中；D 与"sharp knife"建议相反。故选 D。

5.

【答案】B

【题型】细节理解题

【定位】第⑧段 "grown in soil rich in sulphur ... much more of the defence chemical"

【解析】土壤硫量决定刺激程度。A、C、D 均错。故选 B。